

A Test of the International Personality Item Pool Representation of the Revised NEO Personality Inventory and Development of a 120-Item IPIP-Based Measure of the Five-Factor Model

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There has been a substantial increase in the use of personality assessment measures constructed using items from the International Personality Item Pool (IPIP) such as the 300-item IPIP-NEO (Goldberg, 1999), a representation of the Revised NEO Personality Inventory (NEO PI-R; Costa & McCrae, 1992). The IPIP-NEO is free to use and can be modified to accommodate its users' needs. Despite the substantial interest in this measure, there is still a dearth of data demonstrating its convergence with the NEO PI-R. The present study represents an investigation of the reliability and validity of scores on the IPIP-NEO. Additionally, we used item response theory (IRT) methodology to create a 120-item version of the IPIP-NEO. Using an undergraduate sample ($n = 359$), we examined the reliability, as well as the convergent and criterion validity, of scores from the 300-item IPIP-NEO, a previously constructed 120-item version of the IPIP-NEO (Johnson, 2011), and the newly created IRT-based IPIP-120 in comparison to the NEO PI-R across a range of outcomes. Scores from all 3 IPIP measures demonstrated strong reliability and convergence with the NEO PI-R and a high degree of similarity with regard to their correlational profiles across the criterion variables ($r_{ICC} = .983, .972$, and $.976$, respectively). The replicability of these findings was then tested in a community sample ($n = 757$), and the results closely mirrored the findings from Sample 1. These results provide support for the use of the IPIP-NEO and both 120-item IPIP-NEO measures as assessment tools for measurement of the five-factor model.

Keywords: five-factor model, assessment, brief measures, personality

Interest in the science of personality has increased greatly in recent history as scholars recognize personality's important role within and across multiple research domains including education (e.g., Poropat, 2009), work (e.g., Judge, Heller, & Mount, 2002), and romance (e.g., Bouchard, Lussier, & Sabourin, 1999), as well as physical (e.g., Marshall, Wortman, Vickers, Kusulas, & Hervig, 1994) and mental health (e.g., Lahey, 2009). Although many trait models of personality exist (e.g., Eysenck, 1967; Tellegen, 1985) and evidence suggests that these models can be integrated in meaningful ways (e.g., DeYoung, Weisberg, Quilty, & Peterson, 2013; Judge, Rodell, Klinger, Simon, & Crawford, 2013; Markon, Krueger, & Watson, 2005), the five-factor model (FFM) has become the most widely used personality framework in psychology.

Derived from analyses of natural language, the FFM is a hierarchical model consisting of five higher order domains: Neuroticism, Extraversion, Openness, Agreeableness, and Conscientiousness. Although consensus regarding the lower level facet structure of the FFM domains has not been reached (e.g., Costa & McCrae, 1998; Roberts, Bogg, Walton, Chernyshenko, & Stark, 2004), the

widely used Revised NEO Personality Inventory (NEO PI-R; Costa & McCrae, 1992) provides a structure in which each of the five broader domains is associated with six more-specific facets. Researchers have utilized the FFM to understand the contribution of personality to an array of outcomes, including health, relationship quality, occupational choices, and clinical disorders (e.g., Ozer & Benet-Martinez, 2006). More recently, the new pathological trait model included in Section III of the *Diagnostic and Statistical Manual of Mental Disorders* (5th ed. [DSM-5]; American Psychiatric Association, 2013) has been empirically demonstrated to represent maladaptive variants of the FFM (Gore & Widiger, 2013).

Despite the proliferation of research utilizing the FFM, it has been suggested that progress in how to best assess these constructs has been "dismally slow" (Goldberg, 1999, p. 7) for a variety of reasons. Unfortunately, some of the most well-known, comprehensive personality inventories that provide scores for both higher order domains and lower order facets are copyrighted, proprietary instruments, such as the NEO PI-R (Costa & McCrae, 1992) and the Schedule for Assessment of Nonadaptive and Adaptive Personality (Clark, Simms, Wu, & Casillas, in press). The pay-for-use nature of certain assessments can be cost prohibitive for many researchers, especially given the growing acknowledgment that large sample sizes are needed for adequate statistical power and mitigation of concerns regarding replicability. Another limit of proprietary instruments is that they are infrequently revised due, in part, to the fact that other scientists cannot contribute to their continued refinement, thus reducing the flexibility of these instru-

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ments and the manner in which they are used (e.g., one cannot pull certain items or scales out of these proprietary instruments in order to use a limited selection).

It was these issues that led to the development of several freely available measures of the broader FFM domains. However, numerous studies have demonstrated that the specificity provided by lower order facets has much to offer the study of personality (e.g., Paunonen, 1998; Paunonen & Ashton, 2001). Although there may be circumstances in which scores on higher order traits may cross-validate more successfully than lower level facets (e.g., Gruzza & Goldberg, 2007), assessment-oriented scholars have increasingly called for the use of narrow scales in place of scales measuring only the broader, multidimensional scales (e.g., Oswald & Hough, 2011; Strauss & Smith, 2009). With regard to the FFM, the facets have been shown to discriminate better among individuals with personality disorders (PDs) compared to higher order domains (Bagby, Costa, Widiger, Ryder, & Marshall, 2005), can be used to score *DSM-IV* and *DSM-5* PDs (Miller, 2012), and are viewed by clinicians as more clinically useful than domain scores (Sprock, 2002).

In response to the above obstacles, Goldberg and colleagues initiated an international collaborative effort to develop and continually improve a broad and comprehensive pool of freely available personality items titled the International Personality Item Pool (IPIP; Goldberg, 1999, 2001; Goldberg et al., 2006). These items can be used in any way that scientists see fit, are free to use, and do not require permission for their use or modification. A large number of individual scales and broad measures have been created as proxy measures for various copyrighted, proprietary instruments including the NEO PI-R. An IPIP version of the NEO PI-R was created by modifying an existing item pool (Hendricks, 1997) and administering it to an adult community sample in order to develop 10-item scales for each of the 30 FFM facets. The average coefficient alpha for the scales was .80, and the average convergent correlation with the corresponding facets of the NEO PI-R was .73 (Goldberg, 1999), providing preliminary support for the reliability and validity of scores on the IPIP-NEO.

The IPIP has had great impact in the published literature. The chapter introducing the IPIP (Goldberg, 1999) has been cited 1,665 times, items from the IPIP have been used in 581 published studies, 302 scales have been constructed utilizing IPIP items, and the IPIP has been translated into 61 languages. In spite of the use of the IPIP-NEO in several studies, neither its congruence with the NEO PI-R nor its relation to important clinical outcomes has been widely investigated to date. As noted by Goldberg and colleagues in a recent overview of progress related to the IPIP, "one must worry about the extent to which IPIP measures are equivalent to their parent scales" (Goldberg et al., 2006, p. 93); thus, comparative validity studies of the measures are warranted. Therefore, the first goal of the present study was to provide a test of the reliability and construct validity of scores on the 300-item IPIP-NEO.

A major strength of the IPIP-NEO is that it can provide a freely available, comprehensive assessment of the FFM domains as well as the 30 lower order facets. However, the administration of the full 300-item IPIP-NEO can be prohibitive to researchers with time and/or budget constraints and can result in participant fatigue when included as part of a larger overall assessment battery. Several shorter measures that assess the broader FFM/Big Five domains have been created from the IPIP item pool, including

20-item (Donnellan, Oswald, Baird, & Lucas, 2006), 50-item (Goldberg, 1992), 60-item (Goldberg, 2001), 100-item (Saucier & Goldberg, 2002), and 120-item versions (IPIP-Johnson's Short Form [IPIP-J]; Johnson, 2011). Previous research investigating the reliability and validity of scores on these brief IPIP measures has suggested adequate psychometric properties and convergent validity with existing FFM domains (e.g., Donnellan et al., 2006; Gow, Whiteman, Pattie, & Deary, 2005). Although there are several well-functioning domain-level short forms, only the IPIP-J version yields the 30 facets that are associated with the FFM as assessed by the NEO PI-R. The IPIP-J 120-item version was created using an iterative process that focused primarily on the removal of items with the lowest item-total correlations, followed by replacing those items whose content was judged to be too redundant, did not map on to Costa and McCrae's (1992) NEO PI-R, or assessed material judged to be potentially problematic in terms of legality (see Johnson, 2011). Scores on the facets in the 120-item version demonstrated reasonably good internal consistency (mean $\alpha = .68$) and convergence with the NEO PI-R facets (mean $r = .66$). However, there are no published, peer-reviewed data or detailed scale development information on this measure to date.

Given the utility of a free-to-use, 120-item measure that can assess the five domains and 30 facets associated with the NEO PI-R assessment of the FFM, we sought to also test the validity of scores from the IPIP-J 120-item version and create a brief version of the IPIP-NEO utilizing item response theory (IRT). The full IPIP-NEO (Goldberg, 2001) and the 120-item version (Johnson, 2011) were both constructed utilizing classic test theory methods. IRT confers several advantages toward test construction (e.g., Hattie, Jaeger, & Bond, 1999) and allows for a fine-grained examination of the properties of items and their relationship with latent traits. Therefore, the second goal of this study was to utilize IRT techniques to develop and evaluate a new 120-item facet-level NEO-IPIP. For this new measure, we selected the IPIP-NEO items that had the highest measurement precision, or ability to discriminate between people of different trait levels. IRT is particularly well suited for use in the development of short forms because researchers can ensure that the selected items are generally the most reliable and that individuals are measured well across a wide range of trait levels. It has been noted that IRT has not been harnessed in the past to modify existing personality measures such as the NEO PI-R due to copyright issues, which prohibit researchers from changing or modifying items (Reise & Henson, 2003). As such, the development of a 120-item version of the NEO-IPIP represents an opportunity to use modern measurement techniques to create a brief, well-constructed, open-access measure of the FFM domains and facets.

In the current study, we first used IRT to create a 120-item version of the IPIP-NEO. Second, we tested the convergent validity of scores from the 300-item IPIP-NEO, the IPIP-J 120-item version, and our newly created IRT-based version in relation to the NEO PI-R domains and facets. The convergent validity and reliability of the scores from the IPIP-based measures were also investigated in a large community sample (Goldberg, 2008). Next, given the existing literature utilizing measures of the FFM in relation to a variety of important clinical outcomes, we investigated the relations between scores from the four FFM measures with self-report scores on the *DSM-5* PDs, externalizing behaviors (e.g., substance use) and internalizing symptoms (e.g., anxiety), as

well as in relation to informant reports of the new *DSM-5* Section III pathological personality traits.

Method

Participants and Procedure

Sample 1. Participants were 359 undergraduate students (54% female; mean age = 19.35 years, $SD = 2.0$) recruited from a research participant pool at a large Southeastern university who received research credit in exchange for participation. The majority of participants were Caucasian (82%); of the remaining participants, 10% were Asian, and 6% were African American. Following informed consent, participants completed the questionnaires. Participants also provided e-mail addresses for parents who would be able to provide informant reports. After the session, the parents were contacted via e-mail and asked to complete informant reports of the Personality Inventory for *DSM-5* (PID-5). Parent-reports were completed by 145 individuals. Participants with and without informants did not differ with regard to age or gender, but there were fewer American Indian, Asian American, and African American participants than one would have expected in the informant-report sample ($\chi^2 = 17.73$, $p \leq .001$). Participants with and without informants were also compared across the FFM domains as measured by all four FFM measures; participants with informant reports were higher on Openness across all four measures (d s ranged from .26 to .29).

Sample 2. Participants were 757 adults (56.9% female; 98.4% Caucasian) recruited from lists of homeowners from Eugene-Springfield, Oregon (see Goldberg, 2008, for full details) in 1993 who volunteered to complete questionnaires for at least 5 to 10 years. The current participant sample represents those participants who completed at least two of the last four surveys, representing an 88% retention rate between 1993 and 2003. When they were initially recruited in 1993, participants' ages ranged from 18 to 85 years. In 2005, 3.3% of participants were between ages 30 and 41, 16.1% were between ages 42 and 51, 34.7% were between ages 52 and 61, 21.1% were between ages 62 and 71, 16.6% were between ages 72 and 81, and 8.1% were 82 and older.

Measures Included in Both Samples 1 and 2

International Personality Item Pool—NEO (IPIP-NEO). The IPIP-NEO (Goldberg, 2001) is a 300-item self-report inventory of the FFM of personality that assesses the five broad domains and the six lower order facets of each domain. Alphas and mean interitem correlations (MICs) for the domains and facets are presented in Table 1 for all four FFM measures.

Revised NEO Personality Inventory (NEO PI-R). The NEO PI-R (Costa & McCrae, 1992) is a 240-item self-report measure of the FFM of personality that assesses the five FFM broad personality domains as well as the six lower order facets underlying each dimension.

International Personality Item Pool—Johnson's Short Form (IPIP-J). The 120-item, facet-level short form was developed by Johnson by first removing items with the lowest item-total correlations, then conducting content examination related to near-duplicate items, items that may result in legal problems, and

fidelity to NEO PI-R items (Costa & McCrae, 1992), and ensuring that alphas were acceptable at the domain level.

Item Response Theory-Driven (IRT) Short Form (IPIP-120). The IPIP-120 is a 120-item self-report inventory of the FFM of personality that assesses the five broad domains and the six lower order facets of each domain. All items were selected from the 300-item IPIP-NEO.

Sample 1—Only Measures

Demographic form. A brief demographic questionnaire was administered to all participants assessing race, sex, and age.

Crime and Analogous Behavior scale (CAB; Miller & Lyman, 2003). The CAB is a self-report inventory that assesses a variety of externalizing behaviors. Subscales were created via count scores in which participants received a 1 for every item they endorsed (substance abuse: seven items, $M = 2.22$, $SD = 1.52$; antisocial behavior: nine items, $M = .72$, $SD = 1.03$; intimate partner violence: six items, $M = .43$, $SD = .94$; gambling: six items, $M = 1.52$, $SD = 1.40$; risky sex: four items, $M = .47$, $SD = .76$). All subscales except for intimate partner violence were normally distributed; as such, the intimate partner violence subscale was log transformed.

Patient-Reported Outcomes Measurement Information System (PROMIS)—Emotional Distress—Anxiety, Depression, Anger—Short Forms (Pilkonis et al., 2011). The PROMIS scales are brief self-report questionnaires (i.e., seven items for the Anxiety scale; eight items for the Depression scale) designed to assess the experience of a particular emotion over the past 7 days. Alphas in the present study ranged from .91 to .92.

Structured Clinical Interview for DSM-IV Personality Disorders—Personality Questionnaire (SCID-II/PQ). The SCID-II P/Q (First, Gibbon, Spitzer, Williams, & Benjamin, 1997) is a self-report questionnaire designed to assess the diagnostic criteria for the 10 *DSM-IV/5* PDs. In the present study, coefficient alphas ranged from .36 to .71, with a median of .62.

Personality Inventory for DSM-5 (PID-5; Krueger, Derringer, Markon, Watson, & Skodol, 2012). The PID-5 is a 220-item measure of the 25 personality traits of the *DSM-5* PD trait model. Domain scores were computed by summing the trait scales identified as loading on these domains based on the *DSM-5* website and recent factor analytic data (Krueger et al., 2012); certain facets were included in the summation of multiple domains. In the current study, coefficient alphas for the domains ranged from .86 to .94.

Results

Creation of the IPIP-NEO Short Form

IRT methods were utilized to select the IPIP items most similar to the corresponding NEO PI-R items from each facet. Estimation of IRT parameters was accomplished using the IRTPRO software program (Cai, Thissen, & du Toit, 2011). Samejima's (1969) graded response model (GRM) was used fitted to item responses, and item parameters were estimated using the method of marginal maximum likelihood (MML). Samejima's model was chosen because it is capable of modeling graded scales with multiple response options and has previously been applied successfully in the

Table 1
Coefficient Alphas and MICs for the 4 Five-Factor Model Measures

FFM trait	Alpha								MIC								Overlap
	NEO		300		120		IRT		NEO		300		120		IRT		
	S1	S2	S1	S2	S1	S2	S1	S2	S1	S2	S1	S2	S1	S2	S1	S2	
N	.91	.93	.94	.95	.87	.88	.88	.88	.17	.22	.20	.22	.22	.24	.23	.24	16
N1	.77	.83	.84	.83	.81	.71	.81	.71	.29	.38	.34	.32	.52	.37	.52	.37	4
N2	.77	.79	.88	.88	.85	.77	.86	.80	.30	.33	.43	.42	.58	.45	.60	.49	3
N3	.80	.84	.80	.88	.82	.79	.85	.78	.33	.40	.43	.43	.53	.50	.58	.47	3
N4	.68	.74	.80	.80	.71	.63	.72	.64	.21	.26	.29	.28	.38	.29	.39	.31	1
N5	.66	.72	.76	.77	.71	.69	.73	.72	.20	.24	.23	.25	.38	.36	.41	.39	3
N6	.78	.80	.82	.82	.71	.70	.72	.66	.32	.35	.31	.32	.38	.37	.39	.33	2
E	.90	.89	.91	.92	.89	.84	.90	.85	.17	.15	.18	.16	.25	.18	.28	.19	18
E1	.79	.79	.86	.87	.76	.77	.83	.79	.32	.32	.38	.41	.45	.46	.55	.49	2
E2	.81	.81	.88	.79	.81	.62	.84	.72	.36	.35	.42	.28	.51	.29	.57	.39	3
E3	.81	.79	.63	.84	.85	.76	.85	.76	.35	.32	.30	.34	.59	.44	.59	.44	4
E4	.61	.72	.58	.71	.59	.68	.62	.71	.17	.25	.12	.19	.26	.34	.29	.38	3
E5	.56	.64	.82	.78	.75	.68	.76	.71	.17	.19	.33	.28	.44	.35	.46	.39	3
E6	.75	.80	.82	.81	.76	.72	.71	.65	.29	.34	.32	.30	.45	.40	.39	.32	3
O	.90	.91	.93	.92	.87	.86	.88	.85	.16	.17	.18	.17	.21	.19	.23	.20	16
O1	.79	.82	.87	.83	.82	.72	.82	.72	.33	.36	.41	.32	.54	.39	.54	.39	4
O2	.82	.83	.85	.84	.82	.72	.80	.72	.37	.38	.35	.36	.53	.39	.51	.40	3
O3	.71	.75	.78	.81	.67	.66	.80	.76	.24	.27	.25	.30	.34	.33	.51	.44	1
O4	.60	.63	.79	.77	.70	.66	.79	.72	.16	.17	.27	.24	.37	.33	.48	.39	3
O5	.84	.83	.84	.86	.77	.77	.81	.84	.41	.38	.34	.39	.46	.46	.52	.56	3
O6	.69	.79	.80	.86	.73	.78	.78	.78	.22	.31	.28	.38	.34	.42	.46	.45	2
A	.90	.89	.93	.90	.88	.81	.87	.79	.16	.14	.18	.13	.24	.16	.22	.14	18
A1	.83	.85	.86	.82	.88	.71	.88	.71	.38	.43	.38	.32	.64	.39	.64	.39	4
A2	.73	.76	.80	.75	.68	.63	.71	.53	.24	.28	.29	.25	.34	.30	.38	.23	2
A3	.74	.70	.82	.77	.69	.65	.70	.58	.28	.24	.31	.25	.36	.32	.37	.26	2
A4	.72	.72	.77	.73	.74	.55	.80	.63	.24	.25	.26	.22	.43	.26	.49	.30	4
A5	.81	.75	.82	.77	.80	.64	.80	.64	.35	.27	.31	.25	.47	.30	.49	.30	3
A6	.63	.60	.77	.75	.69	.68	.72	.63	.18	.17	.26	.16	.36	.35	.39	.30	3
C	.93	.91	.94	.92	.89	.85	.89	.84	.21	.19	.20	.23	.27	.20	.25	.18	17
C1	.65	.70	.79	.78	.78	.67	.78	.67	.20	.25	.29	.27	.47	.35	.47	.35	4
C2	.77	.74	.86	.83	.81	.75	.78	.73	.29	.27	.37	.33	.52	.43	.47	.41	3
C3	.67	.65	.79	.71	.67	.49	.76	.51	.21	.22	.28	.20	.37	.20	.46	.21	3
C4	.80	.73	.80	.78	.75	.68	.67	.63	.35	.27	.30	.27	.44	.35	.35	.32	2
C5	.81	.80	.85	.85	.70	.66	.77	.78	.36	.33	.37	.37	.37	.34	.46	.47	1
C6	.82	.71	.84	.76	.83	.70	.83	.70	.36	.24	.34	.24	.55	.37	.55	.37	4
M	.76	.78	.82	.82	.77	.71	.79	.72	.27	.28	.30	.28	.42	.34	.44	.35	

Note. NEO = Revised NEO Personality Inventory; 300 = 300-item International Personality Item Pool-NEO (IPIP-NEO); 120 = Johnson's 120-item IPIP-NEO; IRT = 120-item item response theory-based IPIP-NEO; MIC = mean interitem correlation; FFM = five-factor model; S1 = Sample 1; S2 = Sample 2; Overlap = number of items shared by the two 120-item versions of the IPIP-NEO; N = Neuroticism; N1 = Anxiety; N2 = Angry Hostility; N3 = Depression; N4 = Self-Consciousness; N5 = Impulsiveness; N6 = Vulnerability; E = Extraversion; E1 = Warmth; E2 = Gregariousness; E3 = Assertiveness; E4 = Activity; E5 = Excitement Seeking; E6 = Positive Emotions; O = Openness; O1 = Fantasy; O2 = Aesthetics; O3 = Feelings; O4 = Actions; O5 = Ideas; O6 = Values; A = Agreeableness; A1 = Trust; A2 = Straightforwardness; A3 = Altruism; A4 = Compliance; A5 = Modesty; A6 = Tendermindedness; C = Conscientiousness; C1 = Competence; C2 = Order; C3 = Dutifulness; C4 = Achievement Striving; C5 = Self-Discipline; C6 = Deliberation.

personality domain (see Zickar, 1998). Simulation studies have shown that MML item parameters are well estimated in the GRM with sample sizes of 300 for measures with between 10 and 20 items (Kieftenbeld & Natesan, 2012), a situation closely mirroring our own application (estimating item parameters using $N = 359$ and using between 18 and 12 items at a time for calibrating each facet throughout the process of item elimination). These results are consistent with past research that has shown small bias in estimates for sample sizes of 300 or greater (Lautenschlager, Meade, & Kim, 2010; Reise & Yu, 1990).

The GRM supposes that three main parameters can be used to explain response to graded items: (a) the level of the trait of person j , θ_j ; (b) the level of the trait needed to choose an option over the

option below it, $b_{i,k}$ (i.e., how extreme the items' options are); and, most importantly for the current study, (c) the degree to which the item is capable of discriminating between people of different trait levels, a_i , known as the *discrimination* parameter. Discrimination parameters are analogous to factor loadings and item-total correlations in that they represent the degree to which an item hangs onto the measured trait, θ .

The unidimensional GRM was first fitted to all 10 IPIP-NEO items and all eight NEO PI-R items from each facet simultaneously, resulting in a total of 30 sets of item parameters. Therefore, in each of the IRT analyses, θ represented the trait common to the set of NEO PI-R and IPIP-NEO items for the studied facet. We then selected the four IPIP-NEO items that had the highest dis-

crimination parameters among the 10 IPIP-NEO items, and this procedure was conducted for each of the 30 FFM facets. In addition to considering the discrimination of items, throughout this process we closely considered another important consideration in scale construction using IRT: the test *information* function. Whereas classical and factor analytic psychometric models conceive of reliability as a property of the test and/or item, the IRT conception of reliability, or discrimination, acknowledges that some individuals are measured better by a given item than others. More specifically, responses to an item provide the most information about those who have a trait level similar to the level of the trait implied by the item. Those who are further away from the item on the trait continuum are not as reliably measured by that item. During the construction of the IRT-based IPIP-NEO, we confirmed that information was high across the trait range relative to the level of information for the NEO PI-R (see Figure 1 [NEO PI-R facet of Angry Hostility] for an example). As can be seen in Figure 1, there are no levels of θ for which the information function troughs or dips down significantly, suggesting that a similar level of relative precision was attained using the short and long forms.¹

The final set of the eight NEO PI-R items and the four chosen IPIP items was used to evaluate model-data fit. The root-mean-square error of approximation (RMSEA) of these models ranged from 0.03 to 0.06, which indicated adequate model-data fit (see MacCallum, Browne, & Sugawara, 1996), and suggests that the item parameters were interpretable and that one dimension was measured by these sets of items. These analyses resulted in four-item IPIP scales for each of the 30 facets that consisted only of those IPIP-NEO items that were most representative of the same facet measured by the NEO PI-R.

Reliability and Mean Interitem Correlations of the Four FFM Measures

Coefficient alpha and the MIC were calculated for scores on each scale from the four FFM measures (see Table 1). In Sample 1, for the NEO PI-R domain scores, alphas ranged from .90 to .93,

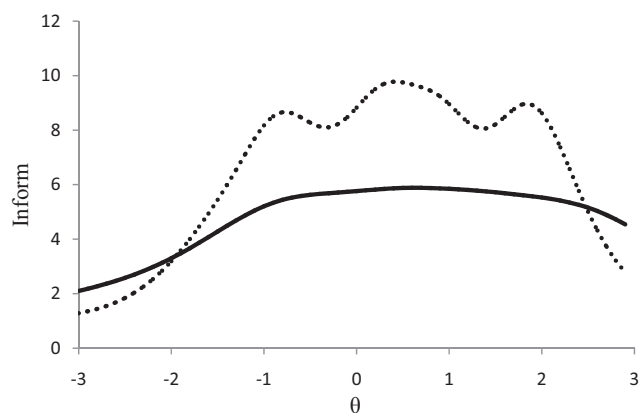


Figure 1. Test information functions for Revised NEO Personality Inventory (solid line) and the final short-form International Personality Item Pool (dotted line) for N2 (Angry Hostility), the second facet of Neuroticism (N).

with a median of .91; alphas for the NEO PI-R facet scores ranged from .56 to .84, with a median of .77. For the IPIP-NEO domain scores, alphas ranged from .91 to .94, with a median of .93; alphas for the facet scores ranged from .58 to .88, with a median of .82. For the IPIP-J domain scores, alphas ranged from .87 to .89, with a median of .88; alphas for the facet scores ranged from .59 to .87, with a median of .75. For the IRT-based IPIP-120 domain scores, alphas ranged from .87 to .90, with a median of .88; the alphas for the facets ranged from .62 to .88, with a median of .78. The mean MICs for scores on the NEO PI-R, IPIP-NEO, IPIP-J, and IPIP-120 were .17, .19, .24, and .24 for the domains and .28, .32, .45, and .48 for the facets, respectively.

In Sample 2, for the NEO PI-R domain scores, alphas ranged from .89 to .93, with a median of .91; alphas for NEO PI-R facet scores ranged from .60 to .85, with a median of .75. For the IPIP-NEO domain scores, alphas ranged from .92 to .95, with a median of .92; alphas for the IPIP-NEO facet scores ranged from .75 to .82, with a median of .77. For the IPIP-J domain scores, alphas ranged from .84 to .88, with a median of .85; alphas for the facet scores ranged from .51 to .80, with a median of .71. For the IRT-based IPIP-120 domain scores, alphas ranged from .84 to .88, with a median of .86; the alphas for the facets ranged from .49 to .71, with a median of .75. The mean MICs for scores on the NEO PI-R, IPIP-NEO, IPIP-J, and IPIP-120 were .17, .18, .19, and .19 for the domains and .30, .29, .37, and .37 for the facets, respectively. The overlap of items on the IPIP-J and IPIP-120 ranged from an overlap of one to four items per facet. The overlap at the domain level ranged from 16 to 18 out of 24 possibly overlapping items, for a total overlap of 85 of 120 items (71%) between the two 120-item IPIP measures.

Convergent Validity: IPIP-NEO Measures With the NEO PI-R

We examined the convergent validity correlations manifested by the NEO PI-R domains and facets scores with the scores on the domains and facets from the three IPIP measures (see Table 2). In order to calculate mean correlations, individual correlations were first transformed using the Fisher's Z transformation before being averaged and transformed back into Pearson correlations. At the domain level for the 300-item IPIP-NEO scores in Sample 1, convergent correlations ranged from .88 to .91, with a median of .89; facet-level convergent correlations ranged from .55 to .85, with a median of .76. The overall mean convergent correlation for the IPIP-NEO scores was .78. At the domain level for the 300-item IPIP-NEO scores in Sample 2, convergent correlations ranged from .83 to .89, with a median of .87; facet-level convergent correlations ranged from .61 to .81, with a median of .72. The overall mean convergent correlation for the IPIP-NEO scores in Sample 2 was .76.

At the domain level for the IPIP-J scores in Sample 1, convergent correlations ranged from .85 to .90, with a median of .87; facet-level convergent correlations ranged from .50 to .84, with a median of .71. The overall mean convergent correlation for the IPIP-J scores was .74. At the domain level for the IPIP-J scores in

¹ Information values should not be directly compared as they are not on the same scale; only relative comparisons of their distribution across θ should be made.

Table 2
Convergent Correlations Among International Personality Item Pool–Based Measures With the Revised NEO Personality Inventory Domains and Facets in Samples 1 and 2

FFM trait	300		120		IRT	
	S1	S2	S1	S2	S1	S2
N	.89	.88	.84	.87	.88	.87
Anxiety	.80	.75	.79	.75	.79	.75
Angry Hostility	.78	.76	.74	.70	.75	.71
Depression	.81	.81	.75	.76	.78	.78
Self-Consciousness	.65	.72	.50	.60	.65	.66
Impulsiveness	.76	.74	.71	.64	.71	.65
Vulnerability	.81	.77	.71	.73	.77	.68
E	.88	.89	.87	.85	.87	.85
Warmth	.75	.77	.68	.68	.69	.72
Gregariousness	.85	.78	.82	.73	.79	.67
Assertiveness	.70	.81	.78	.73	.78	.73
Activity	.55	.71	.51	.62	.54	.63
Excitement Seeking	.68	.67	.64	.59	.64	.60
Positive Emotions	.76	.77	.73	.69	.74	.72
O	.91	.87	.89	.83	.89	.84
Fantasy	.82	.74	.79	.70	.79	.70
Aesthetics	.84	.80	.85	.76	.83	.76
Feelings	.77	.71	.73	.64	.79	.59
Actions	.64	.71	.60	.61	.56	.58
Ideas	.83	.80	.79	.74	.79	.73
Values	.65	.70	.61	.63	.61	.70
A	.88	.83	.85	.76	.87	.78
Trust	.83	.79	.80	.73	.80	.73
Straightforwardness	.70	.64	.61	.54	.72	.60
Altruism	.76	.67	.67	.54	.71	.61
Compliance	.75	.71	.68	.61	.68	.61
Modesty	.80	.72	.76	.64	.77	.63
Tendermindedness	.68	.61	.62	.54	.63	.58
C	.91	.84	.90	.80	.89	.78
Competence	.67	.66	.61	.60	.61	.60
Order	.82	.77	.78	.68	.77	.68
Dutifulness	.63	.61	.58	.53	.60	.49
Achievement	.78	.70	.70	.57	.71	.62
Striving						
Self-Discipline	.78	.76	.77	.71	.72	.66
Deliberation	.82	.69	.78	.61	.78	.61
Mean <i>r</i>	.78	.76	.74	.69	.74	.69
Mean disattenuated <i>r</i>	.97	.94	.94	.85	.95	.92

Note. In Sample 1, correlations $\geq .14$ and $\geq .17$ are significant at $ps \leq .01$ and $.001$, respectively. In Sample 2, correlations $\geq .11$ and $\geq .15$ are significant at $ps \leq .01$ and $.001$, respectively. FFM = five-factor model; 300 = 300-item International Personality Item Pool–NEO (IPIP-NEO); 120 = Johnson's 120-item IPIP-NEO; IRT = 120-item item response theory–based IPIP-NEO; N = Neuroticism; E = Extraversion; O = Openness; A = Agreeableness; C = Conscientiousness; S1 = Sample 1; S2 = Sample 2.

Sample 2, convergent correlations ranged from .76 to .85, with a median of .83; facet-level convergent correlations ranged from .54 to .73, with a median of .64. The overall mean convergent correlation for the IPIP-J scores was .69. At the domain level for the IRT-based IPIP-120 scores in Sample 1, convergent correlations ranged from .87 to .89, with a median of .88; facet-level convergent correlations ranged from .79 to .83, with a median of .72. The overall mean convergent correlation for the IRT-based IPIP-120 scores was .74. At the domain level for the IRT-based IPIP-120 scores in Sample 2, convergent correlations ranged from .78 to .85, with a median of .84; facet-level convergent correlations ranged

from .58 to .73, with a median of .63. The overall mean convergent correlation for the IRT-based IPIP-120 scores was .69. We also calculated the mean disattenuated correlations between the NEO PI-R scores and IPIP scores. In Sample 1, the mean disattenuated convergent correlations were .97, .94, and .95 for the IPIP-NEO, IPIP-J, and IPIP-120, respectively. In Sample 2, the mean disattenuated convergent correlations were .94, .85, and .92 for the IPIP-NEO, IPIP-J, and IPIP-120 scores, respectively.

Discriminant Validity

We next examined the discriminant validity correlations manifested by scores on the NEO PI-R, IPIP-NEO, IPIP-J, and IPIP-120 FFM domains with NEO PI-R FFM domain scores (see Table 3). In Sample 1, NEO PI-R discriminant validity correlations ranged from $-.41$ to $.20$, with an absolute median correlation of $.20$. In Sample 2, NEO PI-R discriminant validity correlations ranged from $-.47$ to $.34$, with an absolute median correlation of $.21$. In Sample 1, IPIP-NEO discriminant validity correlations ranged from $-.39$ to $.21$, with an absolute median correlation of $.17$. In Sample 2, IPIP-NEO discriminant validity correlations ranged from $-.44$ to $.36$, with an absolute median correlation of $.24$. In Sample 1, IPIP-J discriminant validity correlations ranged from $-.41$ to $.26$, with an absolute median correlation of $.17$. In Sample 2, IPIP-J discriminant validity correlations ranged from $-.42$ to $.31$, with an absolute median correlation of $.19$. In Sample 1, IPIP-120 discriminant validity correlations ranged from $-.43$ to $.21$, with an absolute median correlation of $.16$. In Sample 2, IPIP-120 discriminant validity correlations ranged from $-.43$ to $.32$, with an absolute median correlation of $.18$.

Criterion Validity

Relations between the FFM measures and parent-reported PID-5 traits (Sample 1 only). We examined the correlations between the FFM domains, as measured by scores on the four different measures, and informant-reported PID-5 pathological personality domain scores (see Table 4). In general, the domain-level scores from all four FFM measures manifested reasonable convergent and discriminant validity correlations with the informant-reported PID-5 domains. FFM Neuroticism manifested its largest correlations with PID-5 Negative Affectivity, FFM Extraversion manifested its largest (negative) correlations with PID-5 Detachment, FFM Agreeableness manifested its largest (negative) correlations with PID-5 Antagonism, and FFM Conscientiousness manifested its largest correlations with PID-5 Disinhibition. The FFM Openness scores manifested similarly sized correlations with both PID-5 Psychoticism and Disinhibition.

Relations between FFM measures and the DSM-5 PDs (Sample 1 only). We examined the correlations between the scores on the FFM domains as measured by the four different measures and self-report scores on the DSM-5 PDs scores (see Table 5). Across scores on all four FFM measures, Neuroticism demonstrated significant positive correlation with eight of the 10 PDs (all except for Schizoid PD and Antisocial PD). Across scores on all four measures, Extraversion scores consistently demonstrated significant negative relations with Schizotypal, Schizoid, and Avoidant PDs and significant positive relations with Histrionic PD. Across scores on all the four FFM measures, Openness demonstrated significant

Table 3
Discriminant Validity Correlations Among International Personality Item Pool Domains and NEO Domains in Samples 1 and 2

FFM domain	1			2			3			4			5							
	NEO	300	120	IRT	NEO	300	120	IRT	NEO	300	120	IRT	NEO	300	120	IRT				
Sample 1																				
1. N	—	—	—	—	-.30	-.21	-.36	-.34	.01	.04	.07	.10	-.18	-.15	-.14	-.12	-.41	-.39	-.43	
2. E	-.30	-.37	-.41	-.36	—	—	—	—	.20	.20	.11	.10	.14	.14	.20	.15	.20	.21	.18	.20
3. O	.01	-.04	-.09	-.07	.20	.18	.09	.11	—	—	—	—	.13	.20	.26	.20	-.20	-.17	-.15	-.17
4. A	-.18	-.12	-.17	-.16	.14	.05	.07	.05	.13	.15	.12	.06	—	—	—	—	.08	.18	.20	.19
5. C	-.41	-.33	-.27	-.30	.20	.13	.22	.21	-.20	-.16	-.21	-.22	.08	.11	.07	.11	—	—	—	—
Sample 2																				
1. N	—	—	—	—	-.27	-.32	-.35	-.35	-.02	.00	.03	.07	-.21	-.24	-.19	-.18	-.47	-.41	-.43	-.42
2. E	-.27	-.36	-.38	-.35	—	—	—	—	.34	.30	.24	.26	.05	.06	.13	.07	.18	.26	.24	.26
3. O	-.02	-.11	-.12	-.11	.34	.36	.31	.32	—	—	—	—	.02	.10	.19	.11	-.13	-.04	-.04	.01
4. A	-.21	-.25	-.28	-.26	.05	.03	.04	.01	.02	.00	-.02	-.06	—	—	—	—	.15	.17	.22	.17
5. C	-.47	-.44	-.42	-.43	.18	.19	.25	.25	-.13	-.15	-.20	-.21	.15	.12	.10	.11	—	—	—	—

Note. In Sample 1, correlations $\geq .14$ and $\geq .17$ are significant at $ps \leq .01$ and $.001$, respectively. In Sample 2, correlations $\geq .11$ and $\geq .15$ are significant at $ps \leq .01$ and $.001$, respectively. NEO = Revised NEO Personality Inventory; FFM = five-factor model; 300 = 300-item International Personality Item Pool-NEO (IPIP-NEO); 120 = Johnson's 120-item IPIP-NEO; IRT = 120-item item response theory-based IPIP-NEO; N = Neuroticism; E = Extraversion; O = Openness; A = Agreeableness; C = Conscientiousness.

Table 4
Correlations Between Personality Domains and Parent-Reported DSM-5 Personality Disorders in Sample 1

PID-5 domain	Neuroticism				Extraversion				Openness				Agreeableness				Conscientiousness			
	NEO	300	120	IRT	NEO	300	120	IRT	NEO	300	120	IRT	NEO	300	120	IRT	NEO	300	120	IRT
Negative Affectivity	.46	.47	.45	.47	-.08	-.06	-.13	-.13	.02	.05	.04	.08	-.02	-.06	-.06	-.04	-.18	-.18	-.19	-.19
Detachment	.29	.30	.32	.31	-.40	-.40	-.40	-.40	-.09	-.04	-.01	-.19	-.17	-.21	-.21	-.19	-.09	-.10	-.11	-.10
Psychoticism	.28	.23	.18	.18	-.12	-.09	-.18	-.18	.26	.24	.24	.24	-.04	-.09	-.07	-.06	-.23	-.21	-.20	-.19
Antagonism	.16	.15	.14	.18	.07	.07	.06	.06	.00	-.06	-.04	.00	-.30	-.36	-.34	-.35	-.23	-.28	-.28	-.29
Disinhibition	.11	.04	.00	.02	.09	.13	.09	.10	.27	.22	.25	.23	-.15	-.18	-.14	-.18	-.50	-.51	-.51	-.50

Note. Correlations $\geq .22$ and $\geq .27$ are significant at $ps \leq .01$ and $.001$, respectively. DSM-5 = *Diagnostic and Statistical Manual of Mental Disorders* (5th ed.); PID-5 = Personality Inventory for DSM-5; NEO = Revised NEO Personality Inventory; 300 = 300-item International Personality Item Pool-NEO (IPIP-NEO); 120 = Johnson's 120-item IPIP-NEO; IRT = 120-item item response theory-based IPIP-NEO.

Table 5
Correlations Among Five-Factor Model Domains and DSM-5 Personality Disorders in Sample 1

Cluster	Neuroticism			Extraversion			Openness			Agreeableness			Conscientiousness		
	NEO	300	120	NEO	300	120	NEO	300	120	NEO	300	120	NEO	300	120
A															
Paranoid	.40	.40	.39	-.05	-.03	-.05	-.08	-.01	-.01	-.37	-.33	-.31	-.05	-.07	-.10
Schizoid	.05	.08	.12	-.27	-.28	-.26	-.09	-.10	-.07	-.16	-.15	-.19	-.04	-.04	-.05
Schizotypal	.38	.33	.33	-.22	-.22	-.22	.18	.18	.21	-.16	-.16	-.15	-.12	-.10	-.12
B															
Antisocial	.14	.10	.10	-.03	.02	.01	.11	.12	.12	-.31	-.31	-.29	-.24	-.28	-.27
Borderline	.59	.52	.51	-.07	-.06	-.13	.20	.22	.22	-.25	-.22	-.18	-.41	-.39	-.42
Histrionic	.08	.00	-.04	.47	.46	.41	.04	.03	.02	-.24	-.25	-.18	-.11	-.14	-.14
Narcissistic	.25	.17	.16	.12	.12	.12	-.07	-.07	-.05	-.49	-.49	-.48	-.03	-.09	-.10
C															
Avoidant	.50	.53	.53	-.45	-.52	-.54	-.03	-.03	.00	-.02	.00	-.05	-.14	-.16	-.17
Dependent	.39	.40	.37	.01	.01	-.01	.05	.05	.05	.04	.01	.02	-.31	-.30	-.32
OCPD	.23	.23	.24	.07	.06	.09	-.12	-.12	-.16	-.26	-.19	-.19	.26	.27	.23

Note. Correlations $\geq .14$ and $\geq .17$ are significant at $p \leq .01$ and $.001$, respectively. DSM-5 = *Diagnostic and Statistical Manual of Mental Disorders* (5th ed.); NEO = Revised NEO Personality Inventory; 300 = 300-item International Personality Item Pool-NEO (IPIP-NEO); 120 = Johnson's 120-item IPIP-NEO; IRT = 120-item item response theory-based IPIP-NEO; OCPD = obsessive-compulsive personality disorder.

positive relations with Schizotypal and Borderline PDs. Across the four FFM measures, Agreeableness demonstrated significant negative relations with eight of the 10 PDs, including all Cluster A and B PDs, as well as Obsessive-Compulsive PD. Finally, Conscientiousness as measured by scores on the four different FFM measures demonstrated significant negative relations with Antisocial, Borderline, Avoidant, and Dependent PDs, as well as significant positive relations with Obsessive-Compulsive PD.

Relations between the FFM measures and internalizing and externalizing outcomes (Sample 1 only). We examined the correlations between the FFM domains as measured by scores on the four different measures and a range of externalizing and internalizing outcomes (see Table 6). Across scores from the four measures, Neuroticism demonstrated null relations with all five externalizing outcomes and significant positive relations with both internalizing outcomes. Extraversion also demonstrated null relations with all externalizing outcomes but demonstrated significant negative relations with anxiety and depression across the four FFM measures. Openness as measured by scores from the IPIP-NEO, IPIP-J, and IPIP-120 demonstrated small significant negative relations only with gambling, but this correlation was not significant for NEO PI-R Openness. Agreeableness demonstrated significant negative relations with four of the five externalizing outcomes across all four FFM measures, including substance abuse, antisocial behavior, gambling, and number of sexual partners. Conscientiousness manifested significant negative relations with four of the five externalizing outcomes and both internalizing symptoms across all four FFM measures.

Bias and Root-Mean-Square Error

In order to test the overall differences in the correlations manifested by scores on the four IPIP-based FFM measures (IPIP-NEO, IPIP-J, and IPIP-120) with the criterion variables as compared to correlations manifested between the NEO PI-R, bias (i.e., the average difference) and RMSE were calculated. Bias for scores on the IPIP-NEO, IPIP-J, and IPIP-120 compared to the NEO PI-R was small (.0035, .0067, and .0046, respectively). RMSE, compared to the NEO PI-R, was also small across scores on the three measures (.0323, .0429, and .0406, respectively).

Similarity of the Correlational Profiles Manifested by the Four FFM Measures

Finally, the overall similarity of the correlation profiles generated across scores on the four FFM measures across both samples was tested using a double-entry q correlation, an intraclass correlation (ICC) that measures absolute agreement in the correlation profiles (see McCrae, 2008). Scores from all four scales demonstrated significantly similar correlation profiles, with ICCs ranging from .972 to .992. Scores from all three IPIP measures, the NEO-IPIP, NEO-J, and NEO-IRT, demonstrated similar correlation profiles specifically to the NEO PI-R ($r_{ICC} = .983, .972$, and .976, respectively; see Table 7).

Discussion

An increasing interest in the study of personality, as well as the growing acknowledgment of the FFM as an integrative framework

Table 6
Correlations Between the NEO and IPIP-NEO Domains and Outcome Variables in Sample 1

Outcome	Neuroticism			Extraversion			Openness			Agreeableness			Conscientiousness				
	NEO	300	120	IRT	NEO	300	120	IRT	NEO	300	120	IRT	NEO	300	120	IRT	
Externalizing																	
SU	.02	.03	.01	.02	.04	.09	.08	.09	.13	.11	.12	.12	.23	.26	.28	.27	.25
ASB	.08	.03	.07	.06	-.01	.05	.03	.06	.06	.06	.06	.07	.32	.24	.27	.27	.25
IPV	.10	.10	.10	.10	.09	.05	.05	.05	.02	.04	.03	.05	.10	.03	.04	.06	.03
Gam	-.02	-.06	-.05	-.05	-.01	.10	.10	.10	-.09	-.17	.15	.14	.28	.33	.19	.18	.17
Sex	.07	.02	.04	.04	-.06	-.01	-.01	.00	.04	.04	.06	.07	.27	.22	.25	.26	.24
Internalizing																	
Dep	.63	.62	.59	.61	-.28	-.31	-.36	-.33	.07	.08	.12	.13	.14	.35	.36	.37	.39
Anx	.57	.62	.63	.61	-.19	-.22	-.25	-.24	-.03	.02	.05	.04	.05	.15	.20	.22	.23

Note. Correlations $\geq .14$ and $\geq .17$ are significant at $p \leq .01$ and $.001$, respectively. NEO = Revised NEO Personality Inventory; 300 = 300-item International Personality Item Pool-NEO; 120 = Johnson's 120-item item response theory-based IPIP-NEO; SU = substance use; ASB = antisocial behavior; IPV = intimate partner violence; Gam = gambling; Sex = number of sexual partners; Dep = depression; Anx = anxiety.

Table 7
Intraclass Correlations Among the Correlational Profiles
Derived From the Five-Factor Model Measures

Measure	1	2	3	4
1. NEO	—			
2. 300	.983	—		
3. 120	.972	.988	—	
4. IRT	.976	.989	.992	—

Note. NEO = Revised NEO Personality Inventory; 300 = 300-item International Personality Item Pool-NEO (IPIP-NEO); 120 = Johnson's 120-item IPIP-NEO; IRT = 120-item item response theory-based IPIP-NEO.

of personality and personality pathology, has led to a greater need for valid and efficient assessment tools. The NEO PI-R is well validated and widely used in the assessment of the FFM but has limitations, namely, its length and the fact that it is a copyrighted and pay-for-use instrument. Recently, likely in response to increasing demand for open-access and cost-efficient assessment tools, the IPIP-NEO, a free and publically available measure of the FFM, was developed. However, there is a relative dearth of data demonstrating the degree to which the IPIP-NEO mirrors the parent measure from which it was derived, the NEO PI-R. The purpose of the current study was to investigate the reliability and validity of scores from the 300-item IPIP-NEO in comparison to the NEO PI-R, as well as a previously constructed 120-item version of the IPIP-NEO (Johnson, 2011), and to test a newly created IRT-based 120-item version (see Appendix).

Internal Consistency

Scores from the IPIP-NEO, IPIP-J, and IRT-based IPIP-120 manifested good internal consistency as demonstrated by both coefficient alphas and MICs. Scores from the NEO PI-R, IPIP-NEO, IPIP-J, and IPIP-120 demonstrated mean coefficient alphas of .91, .93, .88, and .88 for the domains and .82, .82, .75, and .78 for the facets, respectively. These findings generalized to the second, large community sample used in the current study. The results are also consistent with the original data on the IPIP-NEO (mean $\alpha = .80$; Goldberg, 1999) and preliminary data on the IPIP-J (mean $\alpha = .68$; Johnson, 2011). Scores from both 120-item measures manifested only a small decrement in alpha despite being measured with four to six fewer items per facet, which is important as item number plays a critical role in the calculation of coefficient alpha. To measure internal consistency in a manner that is not contingent upon number of items, we also calculated MICs. The MICs manifested by scores from the NEO PI-R, IPIP-NEO, IPIP-J, and IRT-based IPIP-120 were .17, .19, .24, and .24 for the domains and .28, .31, .45, and .48 for the facets, respectively. This is consistent with preliminary data on the IPIP-J and 300-item IPIP-NEO in which the mean MICs for these measures were .36 and .30, respectively (Johnson, 2011). Clark and Watson (1995) suggested that MICs should fall between .15 and .50 and stated that scales measuring broader constructs (e.g., FFM domains) should manifest lower MICs, whereas scales measuring narrower constructs (e.g., FFM facets) should have higher MICs that might fall in the .40–.50 range. The current data suggest that scores from all four scales demonstrate strong internal consistency, including the

newly created IRT-based IPIP-120 scales and Johnson's 120-item IPIP scales, both of which use 50% and 60% fewer items than the NEO PI-R and IPIP-NEO, respectively.

Convergent Validity

Given that the NEO PI-R is a widely used and well-validated measure of the FFM facets and traits, the convergent validity of scores from three IPIP measures in comparison to this measure was of foremost concern. Data from the current study suggest that scores from the IPIP-NEO, IPIP-J, and IRT-based IPIP-120 demonstrate strong convergent validity in relation to the NEO PI-R, as the mean convergent validity correlation across scores on the 30 facets and five domains was .77, .72, and .72 across the samples. Additionally, the mean disattenuated convergent validity correlations manifested by scores on these measures with the NEO PI-R were .96, .90, and .92 across the samples, demonstrating that scores from the scales correlate approximately as strongly as they can with the parent NEO PI-R scales, given their internal consistencies. Overall, these results provide significant support for the notion that these scales are measuring the same constructs as those measured by the NEO PI-R. The similar relations found across the undergraduate and community samples also provide important support for the external validity of IPIP-based scores.

In general, the current results are largely consistent with preliminary data on the IPIP-NEO (Goldberg, 1999), in which the mean convergent validity correlation of scores from this measure with the NEO PI-R was .73 (.94 after correcting for attenuation). In general, the convergent validity of scores on the three IPIP-based measures with the NEO PI-R was quite substantial and was stronger than the convergence usually found when comparing the NEO PI-R to other measures of the FFM or Big Five. For instance, in a previous study (Gosling, Rentfrow, & Swann, 2003), convergent validity correlations between FFM domains as measured by the Big Five Inventory (John, Donahue, & Kentle, 1991) and the NEO PI-R ranged from .66 to .76, with a median of .68, compared to the range of .88 to .91 ($Mdn = .89$) for the IPIP-NEO, .84 to .90 ($Mdn = .87$) for the IPIP-J, and .87 to .89 ($Mdn = .88$) for the IRT-based IPIP-120.

Discriminant Validity

While the evidence supporting the convergence of scores from the IPIP measures to the NEO PI-R is strong, discriminant validity is another important component of construct validation (Campbell & Fiske, 1959). Discriminant validity was acceptable across scores from all four measures, with absolute median discriminant validity correlations ranging from .16 to .20 in Sample 1 and from .18 to .24 in Sample 2. These discriminant validity correlations in both samples were substantially smaller than the convergent correlations with the IPIP-NEO, IPIP-J, and IRT-based IPIP-120 scores compared to the NEO PI-R (mean $r = .78, .74$, and $.74$, respectively) and are similar to those found in other studies. For instance, in a previous study, the absolute mean FFM discriminant validity correlation was .18 (Gosling et al., 2003). It is also of note that scores from the three IPIP measures demonstrated highly similar discriminant validity coefficients to the NEO PI-R; for instance, in Sample 1, the correlational profiles across the discriminant validity correlations were .964, .937, and .950 for the IPIP-NEO, IPIP-J, and IRT-based IPIP-120, respectively.

Criterion Validity

Although the current data are indicative of strong convergent validity for scores on the IPIP-NEO, IPIP-J, and IPIP-120, scores on different scales can correlate highly but still differ in small but important ways with regard to their correlations with other important criteria. As such, the criterion validity of scores from all four FFM measures was investigated in regard to a range of external criteria. Dimensional models of PD have widespread support, as evidenced by their inclusion in Section III of the *DSM-5*, and there is significant evidence and support for using the FFM as a guiding framework for these kinds of models (e.g., Clark, 2007). In the current study, the FFM domain and facet scores from each measure were investigated in relation to informant reports of the PID-5, the pathological trait measure of the new *DSM-5* Section III PD trait model. In a previous study comparing FFM and the PID-5 domains in an outpatient clinical sample, all of the FFM domains were significantly correlated with their maladaptive counterparts except for Openness, which demonstrated a null relation with its maladaptive counterpart, Psychoticism (Few et al., 2013). In the present study, across all four measures, the FFM domain scores demonstrated significant relations with the maladaptive PID-5 counterpart for all five domains. This is consistent with two factor analytic studies that found that PID-5 Psychoticism facets loaded with FFM Openness (Gore & Widiger, 2013; Thomas et al., 2013). It is noteworthy that in the present study, informant reports of the PID-5 were used given the growing interest in informant reports of personality traits (Vazire, 2006) and evidence that informant reports contribute incremental validity above and beyond self-reports (e.g., Miller, Pilkonis, & Clifton, 2005). The different methodologies used to assess the FFM measures versus the PID-5 likely affected (i.e., decreased) the size of the correlations found between the self-report FFM scales and the informant-report PID-5 scales. Despite these differences, the mean convergent validity correlation across the two methodologies (self vs. informant) and scores from the four FFM measures was .39, suggesting significant agreement across the different models and raters.

Although this dimensional PD model was included in the *DSM-5* as an alternative model in Section III, the *DSM-IV* categorical PDs were retained as the current official diagnostic categories in the *DSM-5*. As such, FFM domain scores, as assessed by the four different measures, were also investigated in relation these 10 official PDs. Consistent with meta-analytic findings regarding the relation between FFM domains and PDs (Saulsman & Page, 2004), Neuroticism and Agreeableness emerged as the domains most consistently related to the *DSM-5* PDs, with Neuroticism consistently positively correlated with multiple PDs and Agreeableness consistently negatively correlated with multiple PDs. These findings were consistent across scores on the NEO PI-R, IPIP-NEO, IPIP-J, and IPIP-120.

Finally, a variety of externalizing and internalizing outcomes were also used in the present study as criterion variables. Meta-analytic evidence suggests a strong relation between Neuroticism and depression and anxiety symptomatology (Kotov, Gamez, Schmidt, & Watson, 2010); this was reflected in the present study in the strong association between Neuroticism scores and depression and anxiety across all four FFM measures. The extant empirical literature has also demonstrated significant relations between both Agreeableness and Conscientiousness and externalizing out-

comes (e.g., Jones, Miller, & Lynam, 2011; Miller, Lynam, & Jones, 2008); in the present study, Agreeableness and Conscientiousness scores were both significantly negatively related to substance abuse, antisocial behaviors, gambling, and number of sex partners across all four measures of the FFM.

Similarity Across the Measures

In order to quantify the similarity of the patterns of correlations for the IPIP-NEO, IPIP-J, and the IPIP-120 compared to the NEO PI-R, multiple indices were used including bias, RMSEA, and ICCs. The pattern of correlations manifested by the five FFM domains with the criterion variables was incredibly similar; the ICCs between the IPIP-NEO, IPIP-J, and the IPIP-120 compared to the NEO PI-R were .983, .972, and .976, respectively. Echoing this similarity across FFM scales, bias, or the average difference from the NEO PI-R correlation, was also small (.0035, .0067, and .0046, respectively, for the IPIP-NEO, IPIP-J, and IPIP-120), as was the RMSEA (.0323, .0429, and .0406, respectively). The findings from these various indices suggest that scores on the IPIP-NEO, IPIP-J, and IPIP-120 manifest nearly identical criterion validity when compared with the NEO PI-R. The similarity of these correlations is crucial because, as stated by Goldberg (1999), the most important test of new personality assessment measures is their utility in predicting and understanding human outcomes. The consistency in the findings of the relations between the FFM domains and self- and parent reports of personality pathology and externalizing and internalizing outcomes provides substantial support for the validity of all three of the IPIP measures.

Scores on the three IPIP-based measures were not only similar to the NEO PI-R scores but were also quite similar to one another as demonstrated by ICCs that ranged from .988 to .992 when comparing scores on the three IPIP measures. This is notable given that both the IPIP-J and IPIP-120 represent an almost two-thirds reduction in items compared to the IPIP-NEO. The overall similarity of scores on the three IPIP measures provides strong support for the use of the IPIP-120 or IPIP-J as a substitute for the much longer NEO PI-R and IPIP-NEO. Given the acknowledgment of the specificity provided by the facet-level data and recent calls for the use of narrow scales (e.g., Oswald & Hough, 2011; Strauss & Smith, 2009), both 120-item IPIP measures provide a viable alternative to the common strategy of using a 50- or 100-item FFM measure that provides domain-level scores only.

It is worth noting that despite the use of IRT in construction of the IPIP-120, scores on this measure and the IPIP-J demonstrated approximately equal convergence and construct validity with regard to the NEO PI-R. It is not surprising that the two short forms demonstrate similar patterns of results, as they overlap substantially with regard to the number of shared items (71%). These results demonstrate that scales created using classical test theory can align closely with those created using more modern IRT construction techniques. The convergence between these two approaches may be particularly likely in certain cases such as on the measurement of normal traits, as opposed to pathological traits in which the range of the latent trait may be more variable. It is of note that our 120-item measure was built purely around psychometric equivalence with the NEO PI-R and had clear psychometric justification for item inclusion and elimination based on sample-independent IRT parameter estimates. The Johnson measure was

built to be equivalent to the 300-item IPIP-NEO (as opposed to the NEO PI-R), was built with classic test statistics that are sample dependent, and also included nonpsychometric considerations when deciding on an item's inclusion (i.e., legal issues). Regardless of these differences in approaches to test construction, the results from the present study suggest that the scores on all three IPIP measures demonstrate strong convergence with NEO PI-R scores and similar patterns of correlations with other important criteria. It is notable that of the 30 FFM scales, only 13 are labeled identically between the NEO PI-R and IPIP-NEO. Given that the mean disattenuated convergent validity correlation between scores on the IPIP-NEO and the NEO PI-R is .97 and the two demonstrate a highly similar pattern of relation with a wide range of outcomes ($r_{ICC} = .987$), the findings of the present study suggest that these differences are likely nominal only.

Limitations and Conclusions

One limitation of the current results is the use of an undergraduate sample for the construction (IRT-based IPIP-NEO) and partial evaluation of the three IPIP measures, which may have impacted the generalizability of these findings or led to a restriction of range for certain variables (e.g., externalizing behaviors or internalizing symptoms). However, the goal of the current study was to compare scores on four measures of the FFM; thus, any attenuation of effect sizes should have happened relatively equally across measures. Additionally, the three IPIP measures performed rather similarly when tested in a larger, community sample, providing initial support for the external validity of the scores from these IPIP measures. Another limitation is that most of the criterion measures were assessed using self-report measures, with the exception of the informant reports collected for the *DSM-5* personality trait model. Again, the reliance on mostly monomethod comparisons is not a particularly salient concern for this study as the goal was to compare the performance of the four measures and thus any increases in effect sizes that might be due to common method variance should affect all four measures of the FFM equally. Additionally, given the comparison of our 120-item IPIP measure to the Johnson (2011) 120-item measure, it is of note that our measure was developed and validated in the same college-student sample, which would provide our measure with an advantage compared to the Johnson measure, which was developed in a large Internet-based sample. However, the cross-validation of both measures in a large community sample suggests that these differences did not have a large or meaningful effect on their performance in Sample 2.

The IPIP represents an exciting opportunity to advance research surrounding the science of personality. The present study demonstrates that scores from the 300-item IPIP-NEO, Johnson's 120-item IPIP measure, and the newly created IRT-based IPIP-120 manifest good reliability, substantial convergence with the NEO PI-R, and strong criterion validity across two samples, suggesting that all three are promising assessment tools for the FFM. The open-access nature of these measures, the ability to use them online (and change them as needed), and their comprehensive coverage of both domains and facets will allow researchers to use these measures in a flexible manner and in the service of collecting large samples. Given the substantial time savings, we believe that both the newly created IPIP-120 and Johnson's 120-item IPIP

measures are particularly promising assessment tools that will prove to have great utility in continuing the substantial advances being made in personality-related scientific endeavors.

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Appendix

IRT-Based IPIP-120 Items

The number in parentheses following the item indicates the corresponding item number in the 300-item IPIP-NEO. An R after the item number indicates that the item is reverse scored.

Neuroticism (N) Facets

N1: Anxiety

- Worry about things. (1)
- Fear for the worst. (31)
- Am afraid of many things. (61)
- Get stressed out easily. (91)

N2: Anger

- Get angry easily. (6)
- Get irritated easily. (36)
- Lose my temper. (126)
- Rarely get irritated. (156R)

N3: Depression

- Often feel blue. (11)
- Dislike myself. (41)
- Am often down in the dumps. (71)
- Have a low opinion of myself. (101)

N4: Self-Consciousness

- Find it difficult to approach others. (76)
- Am easily intimidated. (16)
- Am not embarrassed easily. (196R)
- Am able to stand up for myself. (286R)

N5: Immoderation

- Often eat too much. (21)
- Go on binges. (111)
- Rarely overindulge. (171R)
- Am able to control my cravings. (231R)

N6: Vulnerability

- Feel that I'm unable to deal with things. (86)
- Remain calm under pressure. (176R)
- Know how to cope. (236R)
- Am calm even in tense situations. (296R)

Extraversion (E) Facets

E1: Friendliness

- Make friends easily. (2)
 - Warm up quickly to others. (32)
 - Feel comfortable around people. (62)
 - Act comfortably with others. (92)
- ##### *E2: Gregariousness*
- Love large parties. (7)
 - Talk to a lot of different people at parties. (37)

Don't like crowded events. (217R)

Avoid crowds. (247R)

E3: Assertiveness

- Take charge. (12)
- Try to lead others. (42)
- Take control of things. (132)
- Wait for others to lead the way. (162R)

E4: Activity Level

- Am always busy. (17)
- Am always on the go. (47)
- Do a lot in my spare time. (77)
- Can manage many things at the same time. (107)

E5: Excitement Seeking

- Love excitement. (22)
- Seek adventure. (52)
- Love action. (82)
- Enjoy being reckless. (142)

E6: Cheerfulness

- Radiate joy. (27)
- Have a lot of fun. (57)
- Love life. (147)
- Laugh aloud. (207)

Openness (O) Facets

O1: Imagination

- Have a vivid imagination. (3)
- Enjoy wild flights of fantasy. (33)
- Love to daydream. (63)
- Like to get lost in thought. (93)

O2: Artistic Interests

- See beauty in things that others might not notice. (68)
- Do not like art. (158R)
- Do not like poetry. (188R)
- Do not enjoy going to art museums. (218R)

O3: Emotionality

- Experience my emotions intensely. (13)
- Seldom get emotional. (163R)
- Am not easily affected by my emotions. (193R)
- Experience very few emotional highs and lows. (253R)

O4: Adventurousness

- Prefer to stick with things that I know. (138R)
- Dislike changes. (168R)
- Don't like the idea of change. (198R)
- Am attached to conventional ways. (288R)

(Appendix continues)

O5: Intellect

Am not interested in abstract ideas. (173R)
 Avoid philosophical discussions. (203R)
 Have difficulty understanding abstract ideas. (233R)
 Am not interested in theoretical discussions. (263R)

O6: Liberalism

Tend to vote for liberal political candidates. (28)
 Believe in one true religion. (118R)
 Tend to vote for conservative political candidates. (148R)
 Like to stand during the national anthem. (298R)

Agreeableness (A) Facets*A1: Trust*

Trust others. (4)
 Believe that others have good intentions. (34)
 Trust what people say. (64)
 Distrust people. (184R)

A2: Morality

Use flattery to get ahead. (69R)
 Know how to get around the rules. (129R)
 Cheat to get ahead. (159R)
 Take advantage of others. (249R)

A3: Altruism

Make people feel welcome. (14)
 Love to help others. (74)
 Am concerned about others. (104)
 Turn my back on others. (254R)

A4: Cooperation

Love a good fight. (169R)
 Yell at people. (199R)
 Insult people. (229R)
 Get back at others. (259R)

A5: Modesty

Believe that I am better than others. (144R)
 Think highly of myself. (174R)
 Have a high opinion of myself. (204R)
 Make myself the center of attention. (294R)

A6: Sympathy

Sympathize with the homeless. (29)

Feel sympathy for those who are worse off than myself. (59)
 Suffer from others' sorrows. (119)
 Am not interested in other people's problems. (149R)

Conscientiousness (C) Facets*C1: Self-Efficacy*

Complete tasks successfully. (5)
 Excel in what I do. (35)
 Handle tasks smoothly. (65)
 Know how to get things done. (155)

C2: Orderliness

Like order. (10)
 Like to tidy up. (40)
 Leave a mess in my room. (190R)
 Leave my belongings around. (220R)

C3: Dutifulness

Keep my promises. (45)
 Tell the truth. (105)
 Break my promises. (195R)
 Get others to do my duties. (225R)

C4: Achievement Striving

Work hard. (50)
 Do more than what's expected of me. (140)
 Set high standards for myself and others. (170)
 Am not highly motivated to succeed. (230R)

C5: Self-Discipline

Start tasks right away. (85)
 Find it difficult to get down to work. (175R)
 Need a push to get started. (235R)
 Have difficulty starting tasks. (265R)

C6: Cautiousness

Jump into things without thinking. (120R)
 Make rash decisions. (150R)
 Rush into things. (210R)
 Act without thinking. (270R)

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